Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is: The logs show that: The IP address 203.0.113.0 sends TCP SYN packets to the web server at 192.0.2.1 on port 443. HTTP 504 Gateway Time-out error - log entry 77 and TCP connection resets This event could be: A SYN flood denial-of-service attack - from a single observed IP address

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. SYN - the client sends a SYN packet to request a connection 2. SYN-ACK - the server responds with a SYN-ACK packet and prepares to establish the connection 3. ACK - the client acknowledges the response, completing the TCP connection Explain what happens when a malicious actor sends a large number of SYN packets all at once: When an attacker sends too many SYN requests, the server becomes overloaded. It cannot handle all the requests, so the website becomes slow or stops working. Explain what the logs indicate and how that affects the server: The logs show that the IP address 203.0.113.0 sends many SYN requests to the server 192.0.2.1. At first, the website works normally, but later users start getting errors, such as HTTP 504 Gateway Time-out. After some time, the server stops responding to normal users. As a result, employees cannot access the website, which can cause problems for the company and affect its work.